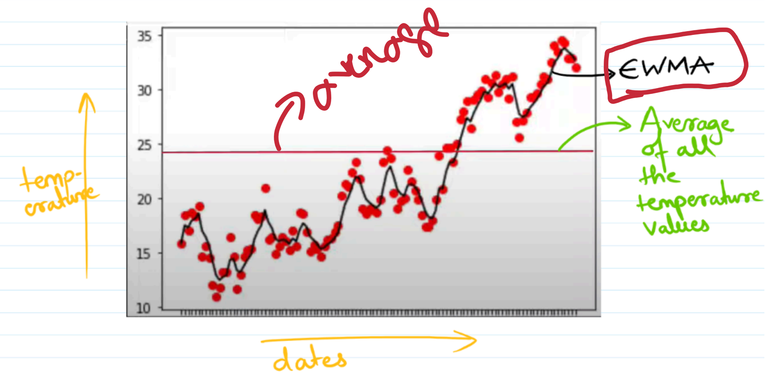
EWMA

[redacted]

[Exponentially weighted moving average]



time series

Stock price prediction

temperature

trend

In india

D1

→ D2 ✓

→ D3 ✓

→ D4 ✓

index	temp(θ)
D1	25
D2	13
D3	17
D4	31
D5	43

$$\boxed{V_t = \beta V_{t-1} + (1-\beta)\theta_t}$$

V_t = EWMA

V_{t-1} = t time of previous

$0 < \beta < 1$

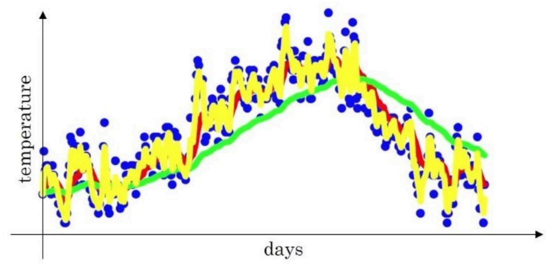
$\beta = 0.9$

$\theta_t =$ t time @

[illegible] data

Exponentially weighted averages

$\boxed{v_t = \beta v_{t-1} + (1-\beta)\theta_t}$ $\beta = 0.9$ 0.98 0.5



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🔹 Role of β in EWMA

Recall:

$$v_t = \beta v_{t-1} + (1-\beta)x_t$$

👉 β controls **how much we remember the past vs. how much we trust the present.**

🔍 Interpretation

- **β close to 1 (e.g., 0.9, 0.99)**
  → Strong memory of past
  → **More smoothing** (less noise)
  → Slower to react to changes
- **β small (e.g., 0.5)**
  → Less memory
  → **Less smoothing** (more responsive)
  → Faster reaction to new values

$v_0 = 0$ | $v_0 = \theta_0$

$v_1 = 0.9 \times v_0 + 0.1 \times 13$

$= 1.3$